Limits Graphically and Numerically

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Estimating limits from a table**

A function g satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \to 3} g(x)$.

Both sides approach 7: $\lim_{x \to 3} g(x) = 7$.

2 **[Foundational]** • **Target: Estimating limits from a table**

Using a table of values with x near 0 (radians), estimate $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).

Values approach 0.50 from both sides, consistent with the exact limit $\frac{1}{2}$.

3 **[Foundational]** • **Target: Reading limits from a graph**

The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ $(x \neq 1)$, state $\lim_{x \to 1} f(x)$.

$\lim_{x \to 1} f(x) = 2$ (even though $f(1)$ itself is undefined — a removable discontinuity).

4 **[Foundational]** • **Target: Reading limits from a graph**

A function h has $\lim_{x \to 2^-} h(x) = 5$ and $\lim_{x \to 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \to 2} h(x)$ and explain whether h is continuous at $x = 2$.

$\lim_{x \to 2} h(x) = 5$, since both one-sided limits agree. But h is NOT continuous at $x = 2$, since $\lim_{x \to 2} h(x) = 5 \neq h(2) = 8$ — this is a removable discontinuity.

5 **[Foundational]** • **Target: One-sided limits and limits that fail to exist**

A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

$\lim_{x \to 1^-} f(x) = 1^2 = 1$. $\lim_{x \to 1^+} f(x) = 3 - 1 = 2$. Since the one-sided limits disagree $(1 \neq 2)$, $\lim_{x \to 1} f(x)$ does not exist.

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6 [Foundational] • Target: Algebraic evaluation

Evaluate each limit by direct substitution or factoring:

a $\lim_{x\to 4}(3x^2-5x+1) \quad = 3(16)-20+1=29$

b $\lim_{x\to 3}\frac{x^2-9}{x-3} \quad = \lim_{x\to 3}(x+3)=6$

c $\lim_{x\to -2}\frac{x^2+5x+6}{x+2} \quad = \lim_{x\to -2}(x+3)=1$

(a) $= 3(16)-20+1=29$; (b) $= \lim_{x\to 3}(x+3)=6$; (c) $= \lim_{x\to -2}(x+3)=1$

7 [Foundational] • Target: Algebraic evaluation

Evaluate $\lim_{x\to 0}\frac{\sqrt{x+9}-3}{x}$ by rationalizing the numerator.

Multiply by $\frac{\sqrt{x+9}+3}{\sqrt{x+9}+3}$: $\frac{x}{x(\sqrt{x+9}+3)}=\frac{1}{\sqrt{x+9}+3}\to\frac{1}{6}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Limits at infinity from a graph

The graph shows $f(x)=\frac{2x}{x+1}$, which has a horizontal asymptote. Use the graph to estimate $\lim_{x\to\infty} f(x)$, then confirm algebraically by dividing numerator and denominator by x.

The graph flattens toward height 2 as $x\to\infty$. Algebraically: $\frac{2x}{x+1}=\frac{2}{1+1/x}\to\frac{2}{1+0}=2$.

9 [Intermediate] • Target: Limits involving infinite discontinuities

For $f(x)=\frac{1}{x-3}$, find $\lim_{x\to 3^-} f(x)$ and $\lim_{x\to 3^+} f(x)$, and state whether $\lim_{x\to 3} f(x)$ exists.

As $x\to 3^-$, $x-3\to 0^-$, so $f(x)\to -\infty$. As $x\to 3^+$, $x-3\to 0^+$, so $f(x)\to +\infty$. Since the one-sided limits are not equal (in fact neither is finite), $\lim_{x\to 3} f(x)$ does not exist.

10 [Intermediate] • Target: Limits involving infinite discontinuities

Evaluate $\lim_{x\to\infty}\frac{3x^2+1}{x^2-4}$, using the fact that the numerator and denominator have the same degree.

For equal-degree rational functions, the limit is the ratio of leading coefficients: $\lim_{x\to\infty}\frac{3x^2+1}{x^2-4}=\frac{3}{1}=3$.

11 [Intermediate] • Target: Estimating limits with a calculator-style table

Construct a mental table of values for $f(x)=\frac{e^x-1}{x}$ at $x=-0.1,\ -0.01, 0.01, 0.1$ (all close to 1), and use it to estimate $\lim_{x\to 0}\frac{e^x-1}{x}$.

The values all cluster near 1 (e.g. $f(0.1)\approx 1.052$, $f(-0.1)\approx 0.952$, both approaching 1 as $x\to 0$): $\lim_{x\to 0}\frac{e^x-1}{x}=1$ (a standard limit related to the derivative of e^x at 0).

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12 **[Intermediate]** • **Target: Limits with piecewise graphical behavior**

A graph shows $f(x) = 2$ for $x < 0$, $f(x) = x + 2$ for $0 \le x < 3$, and $f(x) = 8 - x$ for $x \ge 3$. Find $\lim_{x \to 0} f(x)$ and $\lim_{x \to 3} f(x)$, stating whether each exists.

At $x = 0$: $\lim_{x \to 0^-} f(x) = 2$, $\lim_{x \to 0^+} f(x) = 0 + 2 = 2$: limit exists, equals 2. At $x = 3$: $\lim_{x \to 3^-} f(x) = 3 + 2 = 5$, $\lim_{x \to 3^+} f(x) = 8 - 3 = 5$: limit exists, equals 5.

13 **[Intermediate]** • **Target: Limits with piecewise graphical behavior**

Evaluate $\lim_{x \to 2} \frac{x^2 - 5x + 6}{x^2 - 4}$ by factoring both the numerator and denominator.

$\frac{(x-2)(x-3)}{(x-2)(x+2)} = \frac{x-3}{x+2} \to \frac{2-3}{2+2} = -\frac{1}{4}$.

14 **[Intermediate]** • **Target: Limits involving infinite discontinuities**

Level 3 · AdvancedEvaluate the definite integral using a substitution that transforms the limits: $\int_0^{\sqrt{\pi/2}} x\cos(x^2)e^{\sin(x^2)}\,dx$

Let $u = \sin(x^2)$. Differentiating gives $du = \cos(x^2) \cdot 2x\,dx = 2x\cos(x^2)\,dx \Rightarrow x\cos(x^2)\,dx = \frac{1}{2}du$. Transform the integration limits:Lower limit $x = 0 \Rightarrow u = \sin(0) = 0$.Upper limit $x = \sqrt{\pi/2} \Rightarrow u = \sin((\sqrt{\pi/2})^2) = \sin(\pi/2) = 1$.Substitute into the integral.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Estimating limits with a calculator-style table** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

Level 3 · AdvancedLet $f(x) = x^5 + 3x^3 + 2x - 1$. Find the equation of the tangent line to the graph of $y = f^{-1}(x)$ at $x = 5$.

1. Find a such that $f(a) = 5$: By inspection, $f(1) = 1^5 + 3(1)^3 + 2(1) - 1 = 1 + 3 + 2 - 1 = 5$. So $a = 1$, meaning $f^{-1}(5) = 1$. The point on f^{-1} is $(5, 1)$. 2. Compute $f'(x)$: $f'(x) = 5x^4 + 9x^2 + 2$. At $a = 1$: $f'(1) = 5(1) + 9(1) + 2 = 16$. 3. Slope of f^{-1}: By the inverse derivative formula: $m = (f^{-1})'(5) = \frac{1}{f'(1)} = \frac{1}{16}$. 4. Tangent line: $y - 1 = \frac{1}{16}(x - 5) \Rightarrow \mathbf{y = \frac{1}{16}x + \frac{11}{16}}$.

16 **[Challenging]** • **Target: Limits with piecewise graphical behavior** • **Cross-Lesson: 2.01 The Definition of the Derivative**

Level 1 · FoundationsIf a student completes 24 of the 30 non-calculator questions in Part A with 10 minutes remaining, what pacing adjustment should they make?

The student should not rush into blind guessing. With 10 minutes for 6 questions, they have over 1.6 minutes per question. They should focus on eliminating unreasonable choices and securing answers to the 3-4 most familiar problems before guessing on any remaining unfamiliar ones.

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17 [Challenging] • Target: Estimating limits from a table • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Level 2 · MediumDuring Section I Part B (Calculator), what are the four required calculator capabilities that College Board assesses?

The four required graphing calculator capabilities are: 1. Graphing a function within an arbitrary viewing window; 2. Finding the zeros (roots) of a function; 3. Computing the numerical derivative of a function; 4. Computing the numerical definite integral of a function.

18 [Challenging] • Target: Reading limits from a graph • Cross-Lesson: 3.01 The Chain Rule

Level 2 · MediumThe graph of f' consists of a semicircle of radius 2 centered at the origin, lying above the x-axis. If $f(0) = 5$, find the value of $f(2)$.

By the Fundamental Theorem of Calculus: $f(2) = f(0) + \int_0^2 f'(x)\,dx$. The region under f' from $x = 0$ to $x = 2$ is a quarter-circle of radius 2: $\int_0^2 f'(x)\,dx = \frac{1}{4}\pi(2^2) = \pi$. Therefore, $f(2) = 5 + \pi \approx \mathbf{8.142}$.

19 [Challenging] • Target: One-sided limits and limits that fail to exist • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Level 3 · AdvancedLet $f(x) = x^3 - 3x - 1$. Explain how the Intermediate Value Theorem guarantees a root in $[1, 2]$.

1. $f(x)$ is a polynomial, so it is continuous on $[1, 2]$. 2. $f(1) = 1 - 3 - 1 = -30$. 3. Since 0 lies strictly between $f(1) = -3$ and $f(2) = 1$, by the IVT there must exist at least one $c \in (1, 2)$ such that $f(c) = 0$.

20 [Challenging] • Target: Algebraic evaluation • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Level 1 · FoundationsEvaluate the limit $\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 2x - 3}$.

Direct substitution yields $0/0$. Factor numerator and denominator:
$\lim_{x \to 3} \frac{(x-3)(x+3)}{(x-3)(x+1)} = \lim_{x \to 3} \frac{x+3}{x+1} = \frac{3+3}{3+1} = \frac{6}{4} = \mathbf{\frac{3}{2}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Estimating limits from a table

Evaluate $\lim_{x \to 0} \frac{(x)(x-1)}{x}$ by factoring and cancelling.

a. -3

b. 1

c. -1 ✓

d. 3

Cancel the common factor: $\frac{(x)(x-1)}{x} = x - 1$ for $x \neq 0$, so the limit is $0 - (1) = -1$.

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22 [Foundational] • Target: Reading limits from a graph

Consider the function $f(x)$ defined piecewise as: $f(x) = \begin{cases} \frac{x^2-9}{x-3}, \text{ if } xeq3; k, \text{ if } x = 3 \end{cases}$ For the function $f(x)$ to be continuous at $x = 3$, the value of k must be such that the limit of $f(x)$ as x approaches 3 exists and is equal to $f(3)$. Which of the following statements is true regarding the continuity of $f(x)$ at $x = 3$ if k is set to satisfy this condition, and what is the value of that limit?

a. The function is continuous at $x = 3$ and the limit is 6. ✓

b. The function is continuous at $x = 3$ and the limit is 9.

c. The function is not continuous at $x = 3$ regardless of the value of k.

d. The function is continuous at $x = 3$ and the limit is 3.

For $f(x)$ to be continuous at $x = 3$, the limit of $f(x)$ as x approaches 3 must exist and be equal to $f(3)$. First, we evaluate the limit of $f(x)$ as $x \to 3$: $\lim_{x\to3} \frac{x^2-9}{x-3}$. This is an indeterminate form $\frac{0}{0}$, so we can factor the numerator: $\lim_{x\to3} \frac{(x-3)(x+3)}{x-3}$. For $xeq3$, we can cancel the $(x - 3)$ terms: $\lim_{x\to3} (x + 3)$. Substituting $x = 3$ into the simplified expression gives $3 + 3 = 6$. For contin.

23 [Foundational] • Target: One-sided limits and limits that fail to exist

The function $f(x)$ is defined piecewise as: $f(x) = \begin{cases} ax^2 + 1, , x < 2; 5x - b, , x \geq 2 \end{cases}$ If $f(x)$ is continuous at $x = 2$, and the limit of $f(x)$ as $x \to 2$ is 9, find the values of a and b.

a. a = 2, b = 1

b. a = 1, b = 1

c. a = 2, b = -1

d. a = 1, b = -1 ✓

For $f(x)$ to be continuous at $x = 2$, the left-hand limit and the right-hand limit must be equal to the function value at $x = 2$. Given that the limit of $f(x)$ as $x \to 2$ is 9, we have: 1. Left-hand limit: $\lim_{x\to2^-} f(x) = \lim_{x\to2^-} (ax^2 + 1) = a(2)^2 + 1 = 4a + 1$ 2. Right-hand limit: $\lim_{x\to2^+} f(x) = \lim_{x\to2^+} (5x - b) = 5(2) - b = 10 - b$ 3. Function value at $x = 2$: $f(2) = 5(2) - b = 10 - b$ Since the limit as $x \to 2$ is 9.

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24 [Foundational] • Target: Algebraic evaluation

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.

a. **4** ✓

b. 6

c. 5

d. 7

The function $f(x)$ is defined piecewise. For $0 \le t \le 2$, $f(t)$ is the upper part of a circle centered at the origin with radius 2, so $f(t) = \sqrt{4 - t^2}$. For $2 < t \le 4$, $f(t)$ is a line segment connecting $(2, 4)$ and $(4, 0)$. The equation of this line can be found using the two-point form: $y - 0 = \frac{4-0}{2-4}(x - 4)$, which simplifies to $y = -2(x - 4)$, so $f(t) = -2(t - 4)$. To find $A(3)$, we need to calculate the definite integral $\int_0^3 f(t)\,dt$. We split this integral i.

25 [Foundational] • Target: Limits at infinity from a graph

The graph of the derivative of a function $f(x)$, denoted by $f'(x)$, is shown in the figure above. The function $f(x)$ is continuous, and it is known that $f(1) = 5$. Determine the value of $f(5)$.

a. 11

b. **14** ✓

c. 17

d. 20

The Fundamental Theorem of Calculus states that $\int_a^b f'(x)\,dx = f(b) - f(a)$. Rearranging this, we get $f(b) = f(a) + \int_a^b f'(x)\,dx$. In this problem, we are given $f(1) = 5$ and we need to find $f(5)$. Thus, we have $a = 1$ and $b = 5$. We need to calculate the definite integral $\int_1^5 f'(x)\,dx$, which represents the area under the curve of $f'(x)$ from $x = 1$ to $x = 5$. From the graph, $f'(x)$ is a straight line. The area from $x = 1$ to $x = 5$ can be split into two parts: a triangle a.

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26 [Foundational] • Target: Limits involving infinite discontinuities

The graph of the derivative of a function $f(x)$ is shown below. Given that $f(2) = 5$, determine the value of $f(6)$.

a. 13

b. **18** ✓

c. 23

d. 28

The graph shows $f'(x)$. By the Fundamental Theorem of Calculus, $f(6) - f(2) = \int_2^6 f'(x)\,dx$. The integral $\int_2^6 f'(x)\,dx$ represents the area under the curve of $f'(x)$ from $x = 2$ to $x = 6$. This area is a trapezoid with parallel sides of length 4 (from $x = 2$ to $x = 4$) and 6 (from $x = 4$ to $x = 6$), and a height of 2 (the difference in x-values from 2 to 4 and 4 to 6, but looking at the graph the height from y=1 to y=5 is 4 and from y=1 to y=6 is 5, the vertical distance represents t.

27 [Foundational] • Target: Estimating limits with a calculator-style table

Consider a function $f(x)$ whose derivative is given by $f'(x) = 6x^2 - 12x + 6$. The graph of the function $g(x) = x^3 - 6x^2 + 9x - 2$ is shown below. If $f(x)$ and $g(x)$ have the same tangent line at $x = 1$, determine the value of $f''(0)$.

a. -6 ✓

b. -12

c. 0

d. 6

First, let's find the derivative of $g(x)$: $g'(x) = 3x^2 - 12x + 9$. At $x = 1$, the slope of the tangent line to $g(x)$ is $g'(1) = 3(1)^2 - 12(1) + 9 = 3 - 12 + 9 = 0$. The problem states that $f(x)$ and $g(x)$ have the same tangent line at $x = 1$. This means $f'(1) = g'(1) = 0$. We are given $f'(x) = 6x^2 - 12x + 6$. Let's check: $f'(1) = 6(1)^2 - 12(1) + 6 = 6 - 12 + 6 = 0$. This confirms consistency. To find the value of $f''(0)$, we first need to find the second derivative of $f(x)$. We are given.

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28 [Foundational] • Target: Limits with piecewise graphical behavior

Consider a function $f(x)$ such that its derivative is given by $f'(x) = 3x^2 - 12x + 10$. The graph of the second derivative, $f''(x)$, is shown below. Given that $f(0) = 5$, find the value of $f(2)$.

a. 15

b. **9** ✓

c. 11

d. 21

The derivative of $f'(x)$ is $f''(x)$. From the graph, we can observe that $f''(x)$ is a parabola opening upwards with its vertex at $(2, -2)$. The equation of a parabola with vertex (h, k) is $a(x-h)^2 + k$. Thus, $f''(x) = a(x-2)^2 - 2$. We can also see that the parabola passes through $(0, 10)$, so $10 = a(0-2)^2 - 2$, which gives $12 = 4a$, so $a = 3$. Therefore, $f''(x) = 3(x-2)^2 - 2 = 3(x^2 - 4x + 4) - 2 = 3x^2 - 12x + 12 - 2 = 3x^2 - 12x + 10$. This matches the given $f'(x)$, confirming our observation of.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Estimating limits from a table

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = a$ has the equation $y = 2x - 5$. If $f(a) = -1$ and $f'(a) = 2$, find the value of a.

a. -2

b. -1

c. 1 ✓

d. 2

We are given that the tangent line to the graph of $f(x)$ at $x = a$ has the equation $y = 2x - 5$. The derivative of the function $f(x)$ at $x = a$, denoted as $f'(a)$, represents the slope of the tangent line at that point. The given tangent line equation is in the form $y = mx + c$, where m is the slope. Therefore, the slope of the tangent line is $m = 2$. Thus, we have $f'(a) = 2$. We are also given that $f(a) = -1$. Since the point $(a, f(a))$ lies on the tangent line $y = 2x - 5$, it must satisfy the equation o.

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30 **[Intermediate]** • **Target: Reading limits from a graph**

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. $*If*\int_{0}^{c} f(x) \, dx = k, *whatisthevalueof*\int_{c}^{5} f(x) \, dx*intermsof*k$?

a. 12 + k

b. **12 - k ✓**

c. k - 12

d. k + 2

The total area of the shaded region from $x = 0$ to $x = 5$ is given as 12. This total area can be expressed as the sum of the areas of two sub-regions: the area from $x = 0$ to $x = c$ and the area from $x = c$ to $x = 5$. Mathematically, this can be written using definite integrals: $\int_0^5 f(x)\,dx = \int_0^c f(x)\,dx + \int_c^5 f(x)\,dx$. We are given that the total area is 12, so $\int_0^5 f(x)\,dx = 12$. We are also given that $\int_0^c f(x)\,dx = k$. Substituting these values.

31 **[Intermediate]** • **Target: One-sided limits and limits that fail to exist**

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = 1$ has the equation $y = 2x + k$. Given that $f(1) = 3$, determine the value of k.

a. 1

b. **2 ✓**

c. 3

d. 4

The equation of a tangent line to a function $f(x)$ at a point $(a, f(a))$ is given by $y - f(a) = f'(a)(x - a)$. We are given that the tangent line at $x = 1$ has the equation $y = 2x + k$. The slope of this tangent line is 2, which means $f'(1) = 2$. We are also given that $f(1) = 3$. Substituting these values into the point-slope form of the tangent line equation: $y - 3 = 2(x - 1)$ $y - 3 = 2x - 2$ $y = 2x - 2 + 3$ $y = 2x + 1$ Comparing this to the given tangent line equation $y = 2x + k$, we can see that.

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32 **[Intermediate]** • **Target: Algebraic evaluation**

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.

a. 17

b. **21** ✓

c. 9

d. 11

The area of the shaded region is given by the definite integral of $f'(x)$ from $x = 1$ to $x = 5$. This area is also equal to the change in the function $f(x)$ over this interval: $\int_1^5 f'(x)\,dx = f(5) - f(1)$. The graph shows that the function $f'(x)$ is a constant function, represented by the horizontal line $y = 4$. Therefore, $f'(x) = 4$. The area of the shaded region is given as 16 square units. Using the definite integral: $\int_1^5 4\,dx = [4x]_1^5 = 4(5) - 4(1) = 20 - 4 = 16$. This con.

33 **[Intermediate]** • **Target: Limits at infinity from a graph**

Consider the differential equation $\frac{dy}{dx} = kx - 3$, where k is a constant. The graph of a particular solution $y = f(x)$ to this differential equation is shown below. The tangent line to the graph at $x = 1$ has the equation $y = -2x + 5$. Find the value of k.

a. -3

b. -1 ✓

c. 1

d. 3

The differential equation is $\frac{dy}{dx} = kx - 3$. \\ The given tangent line at $x = 1$ is $y = -2x + 5$. \\ The slope of the tangent line at $x = 1$ is $\frac{dy}{dx}|_{x=1}$. From the equation of the tangent line, the slope is -2. \\ Thus, $\frac{dy}{dx}|_{x=1} = -2$. \\ Substituting $x = 1$ into the differential equation, we get $\frac{dy}{dx}|_{x=1} = k(1) - 3 = k - 3$. \\ Equating the two expressions for the slope at $x = 1$: $k - 3 = -2$. \\ Solving for k: $k = -2 + 3 = 1$. \\ Therefore, the value of k i.

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34 [Intermediate] • Target: Limits involving infinite discontinuities

A differentiable function $f(x)$ is represented by the graph below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 1$ to $x = 4$ is given by the definite integral $\int_1^4 f(x)\,dx$. If $f(x) = -x + 5$ for $1 \le x \le 4$, what is the area of this region?

a. 4.5

b. 6

c. **7.5** ✓

d. 9

The area of the shaded region is given by the definite integral $\int_1^4 (-x + 5)\,dx$. To evaluate this integral, we find the antiderivative of $f(x) = -x + 5$, which is $F(x) = -\frac{x^2}{2} + 5x$. Now we apply the Fundamental Theorem of Calculus: $\int_1^4 (-x + 5)\,dx = F(4) - F(1)$ $F(4) = -\frac{4^2}{2} + 5(4) = -\frac{16}{2} + 20 = -8 + 20 = 12$ $F(1) = -\frac{1^2}{2} + 5(1) = -\frac{1}{2} + 5 = 4.5$ Therefore, the area is $12 - 4.5 = 7.5$. Common errors include incorrectly calculating the anti.

35 [Intermediate] • Target: Estimating limits with a calculator-style table

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = -1$ and a local minimum at $x = 3$. The tangent line to the graph of f at $x = 1$ has the equation $y = -2x + 5$. If $f(0) = 4$ and $f(3) = -5$, find the value of $f'(1)$.

a. -5

b. **-2** ✓

c. 2

d. 5

The derivative of a function at a point represents the slope of the tangent line to the function's graph at that point. We are given that the tangent line to the graph of f at $x = 1$ has the equation $y = -2x + 5$. This equation is in the slope-intercept form $y = mx + b$, where m is the slope. Comparing $y = -2x + 5$ with $y = mx + b$, we see that the slope m is -2. Therefore, the derivative of f at $x = 1$ is $f'(1) = -2$. The information about the local maximum, local minimum, $f(0)$, and $f(3)$ are e.

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36 [Intermediate] • Target: Limits with piecewise graphical behavior

Consider the region bounded by the graph of $f(x) = x^2 + 2$, the x-axis, and the vertical lines $x = 0$ and $x = 3$. Which of the following represents the area of this region?

a. $\int_0^3 (x^2 + 2)\,dx$ ✓

b. $\int_0^3 (x^2 - 2)\,dx$

c. $\int_0^3 x^2\,dx$

d. $\int_0^3 2\,dx$

The area of a region bounded by a curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is given by the definite integral $\int_a^b f(x)\,dx$, provided that $f(x) \geq 0$ over the interval [a, b]. In this case, $f(x) = x^2 + 2$, which is always non-negative for all real x. The region is bounded by $x = 0$ and $x = 3$. Therefore, the area of the region is represented by the definite integral $\int_0^3 (x^2 + 2)\,dx$.

37 [Intermediate] • Target: Estimating limits from a table

The graph of a function $f(x)$ is shown above. What is the instantaneous rate of change of $f(x)$ at $x = 2$?

a. -2

b. **0** ✓

c. 2

d. 4

The instantaneous rate of change of a function at a point is given by its derivative at that point, which corresponds to the slope of the tangent line to the graph of the function at that point. Observing the graph, the function $f(x)$ is a horizontal line segment at $y = 2$ for $0 \leq x \leq 4$. Therefore, the slope of this line segment is 0. Thus, the instantaneous rate of change of $f(x)$ at $x = 2$ is 0.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Reading limits from a graph • Cross-Lesson: 3.01 The Chain Rule

The graph of a function $f(x)$ is shown below. The function is a quadratic with a vertex at $(2, -1)$ and passes through the point $(0, 3)$. Which of the following statements correctly describes the derivative of $f(x)$ at $x = 2$?

a. The derivative is positive.

b. The derivative is negative.

c. **The derivative is zero. ✓**

d. The derivative is undefined.

The graph of the function $f(x)$ is a parabola opening upwards, with its vertex at $(2, -1)$. The vertex of a parabola represents the point where the function reaches its minimum (or maximum) value. At the vertex of a differentiable function, the instantaneous rate of change, which is given by the derivative, is always zero. Therefore, the derivative of $f(x)$ at $x = 2$ is zero.

39 [Challenging] • Target: One-sided limits and limits that fail to exist • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The graph of a differentiable function $y = f(x)$ is shown below. The tangent line to the graph at $x = 2$ has the equation $y = -x + k$. Find the value of k.

a. 4

b. **5 ✓**

c. 6

d. 7

The graph shows that the function $f(x)$ passes through the point $(2, 3)$. Therefore, $f(2) = 3$. The tangent line to the graph at $x = 2$ also passes through the point $(2, f(2))$, which is $(2, 3)$. The equation of the tangent line is given as $y = -x + k$. Substituting the point $(2, 3)$ into this equation, we get $3 = -2 + k$. Solving for k, we find $k = 3 + 2 = 5$.

40 [Challenging] • Target: Algebraic evaluation • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Which of the following represents the coordinates of the vertex of the parabola?

a. **(2, 1) ✓**

b. (1, 0)

c. (3, 0)

d. (2, -1)

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ can be found using the formula for the x-coordinate of the vertex: $x = -b/(2a)$. In this case, $a = -1$ and $b = 4$. So, $x = -4/(2 * -1) = -4/-2 = 2$. To find the y-coordinate, substitute $x = 2$ back into the function: $f(2) = -(2)^2 + 4(2) - 3 = -4 + 8 - 3 = 1$. Therefore, the vertex is at (2, 1).

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41 [Challenging] • Target: Limits at infinity from a graph • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Consider the function $f(x) = x^2 - 5x + 6$. What is the value of $\lim_{x \to 3} f(x)$?

a. 0 ✓

b. 3

c. 6

d. 18

To find the limit of the function $f(x) = x^2 - 5x + 6$ as x approaches 3, we can directly substitute $x = 3$ into the function because it is a polynomial and therefore continuous everywhere. $\lim_{x \to 3} (x^2 - 5x + 6)$ Substitute $x = 3$: $(3)^2 - 5(3) + 6$ $9 - 15 + 6$ $-6 + 6$ 0 Therefore, the limit of the function as x approaches 3 is 0.

42 [Challenging] • Target: Limits involving infinite discontinuities • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

The graph of a differentiable function $f(x)$ is shown below. Which of the following statements about the derivative of $f(x)$, denoted $f'(x)$, at $x = 1$ is true?

a. $f'(1)$ is positive. ✓

b. $f'(1)$ is negative.

c. $f'(1)$ is zero.

d. $f'(1)$ cannot be determined from the graph.

The derivative of a function at a point represents the slope of the tangent line to the graph of the function at that point. Observing the graph at $x = 1$, the function $f(x)$ has a positive slope. Therefore, $f'(1)$ is positive. A negative slope would indicate a decreasing function, and a zero slope would indicate a horizontal tangent line.

43 [Challenging] • Target: Estimating limits with a calculator-style table • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Consider the function $f(x) = x^2 - 5x + 6$. What is the limit of $f(x)$ as x approaches 2?

a. -1 ✓

b. 0

c. 1

d. 2

To find the limit of the polynomial function $f(x) = x^2 - 5x + 6$ as x approaches 2, we can directly substitute $x = 2$ into the function because polynomials are continuous everywhere.
$\lim_{x \to 2} (x^2 - 5x + 6) = (2)^2 - 5(2) + 6 \ = 4 - 10 + 6 \ = 0$ Therefore, the limit is 0.

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44 [Challenging] • Target: Limits with piecewise graphical behavior • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Consider the function $f(x)$ graphed below. What is the limit of $f(x)$ as x approaches 1, i.e., $\lim_{x \to 1} f(x)$?

a. -1

b. **0** ✓

c. 1

d. 2

The limit of a function as x approaches a certain value is the y-value that the function approaches from both the left and the right. Observing the graph, as x gets closer and closer to 1 from both sides, the y-value of the function approaches 0.

45 [Challenging] • Target: Estimating limits from a table • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

A differentiable function f is defined on the interval $[-4, 4]$. The graph of f', the derivative of f, is shown in the figure above. The graph of f' consists of two line segments and a semicircle. (a) Find the value of $f(2) - f(-2)$. Show the setup for your answer. (b) For how many values of c in the interval $(-4, 4)$ does $f'(c) = 0$? Explain your reasoning. (c) The function f has a local maximum at $x = k$. Find the value of k. Explain your reasoning. (d) The function f has an inflection point at $x = p$. Find the value of p. Explain your reasoning.

a. There are no values of c for which $f'(c) = 0$.

b. There is one value of c for which $f'(c) = 0$.

c. **There are two values of c for which $f'(c) = 0$.** ✓

d. There are three values of c for which $f'(c) = 0$.

(a) By the Fundamental Theorem of Calculus, $f(2) - f(-2) = \int_{-2}^{2} f'(x)\,dx$. The region from $x = -2$ to $x = 2$ is a semicircle with radius 2. The area of this semicircle is $\frac{1}{2}\pi(2)^2 = 2\pi$. Therefore, $f(2) - f(-2) = 2\pi$. **(b)** The graph of $f'(x)$ is shown. $f'(c) = 0$ when the graph of f' intersects the x-axis. This occurs at $x = -4$ and $x = 2$. Since we are looking for values of c in the open interval $(-4, 4)$, there is one such value, $c = 2$. **(c)** A local maximum o.

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